Session 2 – Self-bounding learning with random forests

Goal. By the end of the session you should be able to learn the weights of a forest by minimizing a certificate, compare relaxed and exact objectives, relate strength and diversity to the certificates, and select a model with certificates instead of cross-validation.
To reread before the session. Sections 5, 6 and 7.2 of the notes; notebook 4.
Timing (3 hours). Part 2.1 60 min
Part 2.2 45 min
Part 2.3 30 min
Part 2.4 45 min
To hand in at the end of the session. Your notebook, with the table of Part 2.1 with means and standard deviations, and the model-selection table of Part 2.4.

Part 2.1 – Learning the weights (Sections 6 and 7.2)

1. On each dataset, compute the posteriors minimizing the first-order bound (`optimize_fo`) and the tandem bound (`optimize_tnd`). Report the test risk of the corresponding votes, their certificates and the *effective number of trees* $1/\sum_t \rho_t^2$.
2. Produce a table with one row per dataset and, for each posterior (uniform, FO, TND), the mean and standard deviation over the 5 splits of the test risk and of the certificate. For the uniform posterior, report the best of the three certificates: what correction does this choice require?
3. Which posterior gives the best vote? the best certificate? Are the conclusions of notebook 4 confirmed?

Solution

Algorithm 5: Learning the weights of the forest (FO: alternating minimization; TND: L-BFGS on logits + closed-form λ)

Input: OOB statistics $\widehat{L}$, $\widehat{\mathbf{L}}$, n_1, n_2; prior π uniform

$\rho_{\mathrm{FO}} \leftarrow$ Algorithm 3 of the notes: repeat $\rho \propto \pi e^{-\lambda n_1 \widehat{L}}$, $\lambda \leftarrow \lambda^\star(\rho)$;

$\rho_{\mathrm{TND}} \leftarrow$ Algorithm 4 of the notes: repeat $\theta \leftarrow \arg\min_\theta \mathcal{B}_{\mathrm{TND}}(\mathrm{softmax}(\theta), \lambda)$, $\lambda \leftarrow \lambda^\star$;

for $\rho \in \{\pi, \rho_{\mathrm{FO}}, \rho_{\mathrm{TND}}\}$ **do**

  test risk of B_ρ; effective number $1/\sum_t \rho_t^2$;

certificates: FO of ρ_{FO}, TND of ρ_{TND}, and $\min(\mathrm{FO}, \mathrm{TND}, \mathrm{C})$ of π with $\delta/3$ each;

Python: function `part2_3` of `tp_solutions.py`.
Means (standard deviations) over the 5 splits; "eff." is the effective number of trees. For the uniform vote, the certificate is the minimum of the FO, TND and C-bound certificates computed with $\delta/3$ each (union bound, Proposition 6.3), since the best bound is chosen after seeing them.

dataset	uniform test risk	uniform best cert.	FO-optimized test risk	FO-optimized FO cert.	FO-optimized eff.	TND-optimized test risk	TND-optimized TND cert.	TND-optimized eff.
australian	0.141 (0.011)	0.666 (0.006)	0.170 (0.029)	0.509 (0.019)	3.2	0.138 (0.013)	0.772 (0.013)	58.3
balance	0.097 (0.016)	0.540 (0.014)	0.137 (0.025)	0.442 (0.023)	5.6	0.101 (0.020)	0.633 (0.029)	71.8
bupa	0.254 (0.038)	1.000 (0.000)	0.367 (0.066)	0.985 (0.019)	5.0	0.283 (0.049)	1.000 (0.000)	66.8
german	0.242 (0.010)	0.879 (0.011)	0.297 (0.035)	0.761 (0.027)	2.3	0.248 (0.014)	0.998 (0.003)	51.8
heart	0.158 (0.044)	0.964 (0.025)	0.212 (0.040)	0.797 (0.027)	8.0	0.165 (0.040)	1.000 (0.000)	74.4
iono	0.077 (0.026)	0.584 (0.016)	0.079 (0.014)	0.440 (0.020)	7.5	0.075 (0.028)	0.739 (0.022)	76.6
pima	0.249 (0.021)	0.864 (0.023)	0.268 (0.028)	0.765 (0.016)	6.0	0.247 (0.023)	0.999 (0.001)	57.6
sonar	0.216 (0.022)	1.000 (0.000)	0.305 (0.027)	0.897 (0.049)	6.6	0.206 (0.010)	1.000 (0.000)	74.3
wdbc	0.061 (0.014)	0.340 (0.014)	0.070 (0.018)	0.258 (0.024)	10.3	0.060 (0.015)	0.438 (0.018)	72.5
spambase	0.056 (0.005)	0.298 (0.004)	0.096 (0.015)	0.263 (0.007)	1.9	0.054 (0.003)	0.272 (0.005)	46.0
abalone8	0.141 (0.003)	0.507 (0.002)	0.178 (0.030)	0.472 (0.005)	5.3	0.143 (0.002)	0.572 (0.003)	51.3
pageblocks	0.021 (0.002)	0.127 (0.001)	0.034 (0.006)	0.105 (0.002)	3.2	0.022 (0.002)	0.140 (0.004)	46.2
satimage	0.062 (0.002)	0.252 (0.004)	0.094 (0.004)	0.227 (0.005)	2.1	0.061 (0.004)	0.258 (0.006)	47.5
segmentation	0.027 (0.010)	0.216 (0.005)	0.049 (0.013)	0.152 (0.010)	1.6	0.025 (0.009)	0.199 (0.004)	47.8
yeast3	0.046 (0.009)	0.282 (0.008)	0.053 (0.008)	0.220 (0.005)	6.3	0.048 (0.006)	0.325 (0.011)	56.0
mean	0.123	0.568	0.161	0.486		0.125	0.623	

Best vote. The TND-optimized vote is as good as the uniform one (mean 0.125 vs 0.123, differences within one standard deviation on every dataset). The FO-optimized vote is worse on all 15 datasets (only marginally on `iono`; mean 0.161), sometimes by a lot (`bupa`: 0.254 → 0.367, `sonar`: 0.216 → 0.305). The effective number of trees explains why: the FO posterior keeps between 2 and 10 trees, the TND posterior about 50 to 75. Minimizing a first-order bound selects the individually best trees, which are correlated, and destroys the diversity that makes the vote work.
Best certificate. Paradoxically, the smallest certificate is almost always the FO certificate of the FO-optimized posterior (mean 0.486), because it is estimated on more examples and it minimizes precisely this number. But it certifies a worse vote. The TND certificate of the TND-optimized posterior is smaller than (or equal to, when both are vacuous) the TND certificate of the uniform vote on every dataset (e.g. 0.289 → 0.272 on `spambase`), and it becomes the best compromise when the data are large enough. This reproduces the conclusions of notebook 4 and of Section 7.2: a certificate that ignores the correlations between voters should not be used as a learning objective for a vote.

Part 2.2 – Minimizing the kl certificate directly (Section 6.6)

The optimizers of Part 2.1 minimize a *relaxation* of the certificate (PAC-Bayes-λ). Since the certificate is valid for any posterior, we can also minimize the kl-form certificate itself.

1. Using `certificate_and_grad` (value and gradient of $4\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{\mathbf{L}}\rho, \varepsilon(\rho))$ with respect to the softmax logits, Proposition 6.6), implement Adam (300 iterations, step 0.05, starting from the uniform posterior).

2. On `spambase`, `satimage`, `wdbc` and `pageblocks`, compare the uniform posterior, the TND-optimized posterior of Part 2.1 and the posterior found by Adam: certificate, test risk, effective number of trees. Plot the certificate along the iterations.

3. Does the choice of the optimizer, of the step size or of the number of iterations require a union bound? Why?

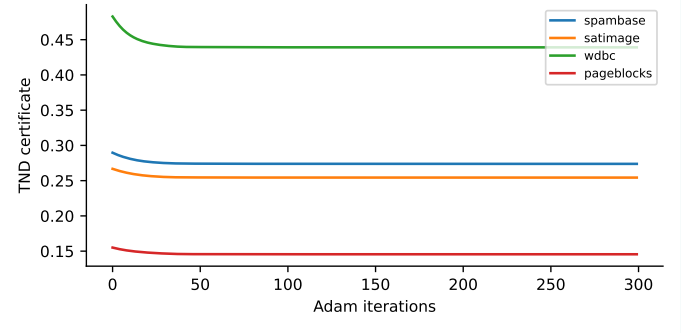
Solution

Algorithm 6: Adam on the kl-form tandem certificate

$\theta \leftarrow \ln \pi$; $u \leftarrow 0$; $v \leftarrow 0$;
for $k = 1, \dots, K$ **do**
  $(\mathcal{B}, g) \leftarrow$ `certificate_and_grad`(θ) ; ▷ $g = \rho \odot (g_\rho - \rho^\top g_\rho)$, g_ρ *from the kl derivatives*
  $u \leftarrow 0.9u + 0.1g$; $v \leftarrow 0.999v + 0.001g^2$;
  $\theta \leftarrow \theta - \eta \frac{u/(1-0.9^k)}{\sqrt{v/(1-0.999^k)}+10^{-8}}$;
return $\rho = \mathrm{softmax}(\theta)$ and its certificate;

Python: function `adam_certificate, partB` of `tp_solutions.py`.

dataset	uniform cert.	uniform test	TND-λ (Part 2.1) cert.	TND-λ test	TND-λ eff.	Adam on the kl certificate cert.	Adam test	Adam eff.
spambase	0.291	0.055	0.274	0.053	47	0.273	0.053	41
satimage	0.268	0.063	0.255	0.064	44	0.255	0.064	39
wdbc	0.470	0.060	0.432	0.058	72	0.424	0.058	42
pageblocks	0.151	0.021	0.140	0.021	46	0.140	0.021	36



2. Both optimizers improve the certificate of the uniform forest by a few points (0.291 → 0.273 on `spambase`, 0.470 → 0.424 on `wdbc`) without changing the test risk of the vote. Minimizing the exact kl certificate is marginally better than minimizing its PAC-Bayes-λ relaxation (at most 0.008 on `wdbc`), which confirms that the relaxation is tight near the optimum; Adam keeps slightly fewer effective trees (36–42 instead of 44–72). The curves show that most of the gain is obtained in the first 50 iterations.
3. No union bound is needed: the optimizer, its hyper-parameters and its stopping time only determine *which* posterior is returned, and the certificate is valid for every posterior simultaneously (Theorem 6.2). One may therefore try several optimizers and keep the best certificate, provided the prior, δ and the form of the bound are unchanged.

Part 2.3 – Strength versus diversity (Section 5)

1. On `spambase` and `german`, learn forests with `max_features` $\in \{1, \sqrt{d}, d/2, d\}$ (3 splits). Report the test Gibbs risk, the test disagreement, the test risk of the vote and the OOB tandem and C-bound certificates.
2. Describe how strength and diversity evolve. Which value gives the best vote? Do the certificates follow the risk of the vote?

Solution

Algorithm 7: Strength–diversity study

for *max_features* $\in \{1, \sqrt{d}, d/2, d\}$, *split* $s = 1, 2, 3$ **do**
  learn the forest; on the test sample compute $R(G_\pi)$, $d_\pi = 2(R(G_\pi) - e_\pi)$ and $R(B_\pi)$;
  compute the OOB TND and C-bound certificates (Part 1.3);

Python: function `part5` of `tp_solutions.py`.

dataset	`max_features`	Gibbs	d	vote	TND cert.	C-bound cert.
spambase	1	0.142	0.192	0.048	0.307	0.413
spambase	$\sqrt{d}$	0.123	0.158	0.055	0.291	0.382
spambase	$d/2$	0.116	0.136	0.060	0.302	0.377
spambase	d	0.114	0.128	0.064	0.309	0.379
german	1	0.362	0.376	0.264	1.000	0.981
german	$\sqrt{d}$	0.346	0.363	0.244	1.000	0.969
german	$d/2$	0.335	0.349	0.234	1.000	0.961
german	d	0.333	0.343	0.242	1.000	0.959

Trying more features per split makes each tree stronger (the Gibbs risk decreases, from 0.142 to 0.114 on `spambase`) but the trees more similar (the disagreement decreases from 0.192 to 0.128). The vote is best with a single feature on `spambase` (0.048) and with $d/2$ features on `german` (0.234): the optimum is a trade-off, as predicted by the C-bound (Figure 35 of the notes). On `spambase` the tandem certificate is minimal at $\sqrt{d}$ (0.291) and the C-bound certificate decreases and then stabilizes: the certificates see the trade-off but not its exact optimum, because their estimation error varies with the configuration. On `german` (700 training examples) the tandem certificate is vacuous and the C-bound certificate hardly informative: with few examples, certificates cannot guide such a fine choice.

Part 2.4 – Self-bounding model selection (Section 6)

1. On `spambase`, `pima`, `wdbc` and `german`, learn $K = 5$ forests with `max_depth` $\in \{1, 2, 4, 8, \text{None}\}$, minimize the tandem bound of each of them and compute its certificate with δ/K. Select the depth with the smallest certificate.
2. Compare with the depth selected by 5-fold cross-validation of the uniform forest on the training set, and with the best depth in hindsight (smallest test risk). Report the test risks of the selected models.
3. Why must the certificates be computed with δ/K? When does this selection rule fail, and what should be done then?

Solution

Algorithm 8: Model selection with certificates versus cross-validation

for $k = 1, \ldots, K$ *(depth* D_k*)* **do**
  forest F_k on the training set; $\rho_k \leftarrow$ `optimize_tnd` with δ/K; $c_k \leftarrow$ TND certificate with δ/K;
  $\text{cv}_k \leftarrow$ 5-fold cross-validation error of the uniform forest of depth D_k;
$k_{\text{cert}} \leftarrow \arg\min_k c_k$ (certified at level $1 - \delta$); $k_{\text{CV}} \leftarrow \arg\min_k \text{cv}_k$ (no certificate);

Python: function `part6` of `tp_solutions.py`.
Selected depths on 3 splits, mean test risk of the selected forest, and mean certificate of the model selected by the certificate:

dataset	depth (cert.)	test risk	depth (CV)	test risk	best	certificate
spambase	None, None, None	0.052	None, None, None	0.052	0.052	0.286
pima	1, 1, 1	0.254	4, None, 4	0.238	0.225	1.000
wdbc	8, 8, 8	0.058	4, 4, 8	0.058	0.055	0.489
german	1, 1, 1	0.294	None, None, None	0.247	0.240	1.000

Why δ/K. Keeping the smallest of K certificates is a choice made with the data; the certificate of the selected model is valid only if all K certificates hold simultaneously, which the union bound guarantees (Proposition 6.3). The price is $\ln 5 \approx 1.6$ nats, negligible here.
When it works. On `spambase` and `wdbc`, where the certificates are informative (0.29 and 0.49), the certificate selects deep trees and reaches the same test risk as cross-validation, without any validation fold and with a guarantee on the selected model.
When it fails. On `pima` and `german`, all the certificates equal 1: they are vacuous and carry no information, and the argmin returns the first candidate (depth 1), which is a poor choice (0.254 vs 0.238 for cross-validation on `pima`, 0.294 vs 0.247 on `german`). A selection rule based on certificates is meaningful only when the certificates are non-vacuous: with small samples, one should fall back on cross-validation (and certify the selected model on a held-out sample, Part 3.2), or use a first-order or data-dependent-prior certificate, which is less loose on small data.